Predicting 30-Day Hospital Readmissions for Diabetic Patients

Machine Learning Analysis Using the UCI Diabetes 130-US Hospitals Dataset

MSDS 422: Practical Machine Learning
Final Project Report

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March 2026

Executive Summary

Hospital readmissions within 30 days of discharge represent a significant and costly challenge in American healthcare, with estimates exceeding $26 billion annually. Under the Centers for Medicare and Medicaid Services (CMS) Hospital Readmissions Reduction Program, hospitals face direct financial penalties for excess readmission rates. For diabetic patients—who face readmission rates between 9 and 14 percent—accurate risk prediction at discharge could enable targeted interventions that reduce both costs and patient suffering.

This project develops and compares four machine learning models for predicting 30-day readmission risk using the UCI Diabetes 130-US Hospitals dataset, which contains 101,766 inpatient encounters across 130 hospitals from 1999 to 2008. After preprocessing—including duplicate removal, invalid discharge filtering, and missing data handling—approximately 69,973 encounters were retained. We engineered 10 new features capturing prior healthcare utilization, medication management patterns, length-of-stay characteristics, and clinical complexity indicators, resulting in a final feature set of 55 variables.

Four models were trained and evaluated using 5-fold stratified cross-validation: Logistic Regression (interpretable baseline), Random Forest (ensemble method), XGBoost (gradient boosting), and a Multi-Layer Perceptron neural network. XGBoost achieved the highest test AUC-ROC of 0.644, followed by Random Forest at 0.633, Logistic Regression at 0.621, and the MLP at 0.611. While these results fall slightly below the published benchmark of 0.667 reported by Emi-Johnson and Nkrumah (2025), our deduplication strategy (retaining only the first encounter per patient) reduced the dataset from 101,766 to 69,973 encounters and shifted the positive class rate from 11.2% to 9.0%, creating a more challenging prediction task with a 10.1:1 imbalance ratio.

The most influential predictors across all models were the number of prior inpatient visits, number of emergency visits, number of diagnoses, time in hospital, and insulin-related medication changes. We recommend deploying the XGBoost model in a clinical risk-flagging system at discharge, targeting the top 20 percent highest-risk patients with enhanced post-discharge interventions. Based on published estimates, this approach could reduce readmissions by 10 to 15 percent, translating to potential savings exceeding $2.6 million annually per hospital.

Problem Statement and Research Objectives

Business Impact

Hospital readmissions impose substantial financial and operational burdens on healthcare systems. Each readmission costs hospitals between $10,000 and $15,000 on average, and since 2012, CMS has penalized hospitals with excessive readmission rates by reducing Medicare reimbursements by up to 3 percent. Diabetic patients face particularly elevated readmission risk due to complex medication regimens, multiple comorbidities, and the need for continuous disease management.

Many readmissions are preventable through targeted interventions such as patient education, medication reconciliation, and timely follow-up care. However, resource constraints prevent hospitals from providing intensive post-discharge support to every patient. Accurate predictive models enable hospitals to allocate limited resources to the patients most likely to benefit, improving outcomes while reducing costs and regulatory penalties.

Research Objectives

This project aims to: (1) develop and compare four machine learning models for predicting 30-day readmission risk in diabetic patients, targeting an AUC-ROC competitive with the published benchmark of 0.667; (2) identify and rank the clinical and demographic risk factors most strongly associated with readmission; (3) design an automated, reproducible end-to-end pipeline from data ingestion through model deployment; and (4) provide actionable, data-driven recommendations for hospital management regarding targeted discharge interventions.

Literature Review

Nine peer-reviewed papers were reviewed, organized into three thematic areas with three papers per team member.

Readmission Risk Factors and Costs

Strack et al. (2014) conducted the original analysis of the UCI Diabetes 130-US Hospitals dataset, demonstrating that HbA1c measurement during hospitalization significantly impacts readmission rates and identifying medication management as a key modifiable risk factor. Their work established the foundational understanding of which clinical variables carry predictive value for this patient population.

Rubin (2015) quantified the economic burden, estimating that each diabetes-related readmission costs upward of $25,000 and that early identification and intervention programs can reduce these costs by 15 to 20 percent. This analysis provided the cost-benefit justification for investing in predictive modeling infrastructure.

Donzé et al. (2016) developed the HOSPITAL score for predicting 30-day readmissions, identifying the number of procedures, sodium levels, and length of stay as strong predictors. Their validated scoring system informed our feature engineering strategy, particularly the creation of procedure intensity and length-of-stay bin features.

Machine Learning Approaches for Clinical Prediction

Hammoudeh et al. (2018) demonstrated that gradient boosting methods outperform logistic regression for clinical prediction tasks involving imbalanced datasets, establishing a precedent for our model selection. Their findings confirmed that tree-based ensemble methods handle sparse clinical features more effectively than linear models.

Rajkomar et al. (2018) showed that deep learning applied to electronic health record data achieves strong results but requires careful feature engineering to realize its potential on structured tabular data. Their work informed our decision to include an MLP neural network as one of our four models while maintaining realistic expectations about its performance on this type of data.

Chen and Guestrin (2016) introduced the XGBoost framework, which provides regularized gradient boosting with built-in handling of sparse data and missing values. XGBoost has since

become the standard approach for tabular clinical prediction tasks, and our implementation leverages its scale_pos_weight parameter for direct class imbalance handling.

Feature Engineering and Imbalanced Learning in Healthcare

Dugan and Shubrook (2020) confirmed that prior healthcare utilization—specifically emergency room visits and prior inpatient stays—ranks among the strongest predictors of diabetic readmission, outperforming demographic variables. This finding directly guided our creation of total prior visits, visit intensity categories, and the high-utilizer flag.

Choi et al. (2016) compared neural network approaches including recurrent architectures and MLPs for EHR data, finding that while neural networks show promise, gradient boosting remains competitive on structured tabular datasets. Their RETAIN model emphasized the importance of interpretability in clinical contexts.

Fernández et al. (2018) provided a comprehensive review of techniques for learning from imbalanced data, recommending the combination of oversampling methods with tree-based models and cost-sensitive learning. Their framework informed our approach to handling the 10.1:1 class imbalance through scale_pos_weight and balanced class weighting.

Exploratory Data Analysis

Dataset Overview

The UCI Diabetes 130-US Hospitals dataset contains 101,766 inpatient encounters across 130 United States hospitals from 1999 to 2008. Each record includes 55 features spanning demographics, admission details, clinical metrics, diagnoses, medications, laboratory results, and prior utilization. After removing duplicate patient encounters (keeping only the first), filtering invalid discharges (deceased and hospice), and dropping high-missingness columns, 69,973 encounters remained for analysis.

Missing Data Analysis

Figure 1 presents the missing data profile across features. Three variables exhibited substantial missingness: weight (97 percent), max_glu_serum (94 percent), and A1Cresult (83 percent). Weight was dropped due to excessive missingness that would render any imputation unreliable. The glucose and HbA1c fields, while technically recorded as missing, represent clinically meaningful information: the absence of these tests indicates they were not ordered, which itself may reflect clinical judgment about the patient's condition. Medical specialty (47 percent missing) was retained with missing values coded as "Unknown," and payer_code (39 percent missing) was dropped.

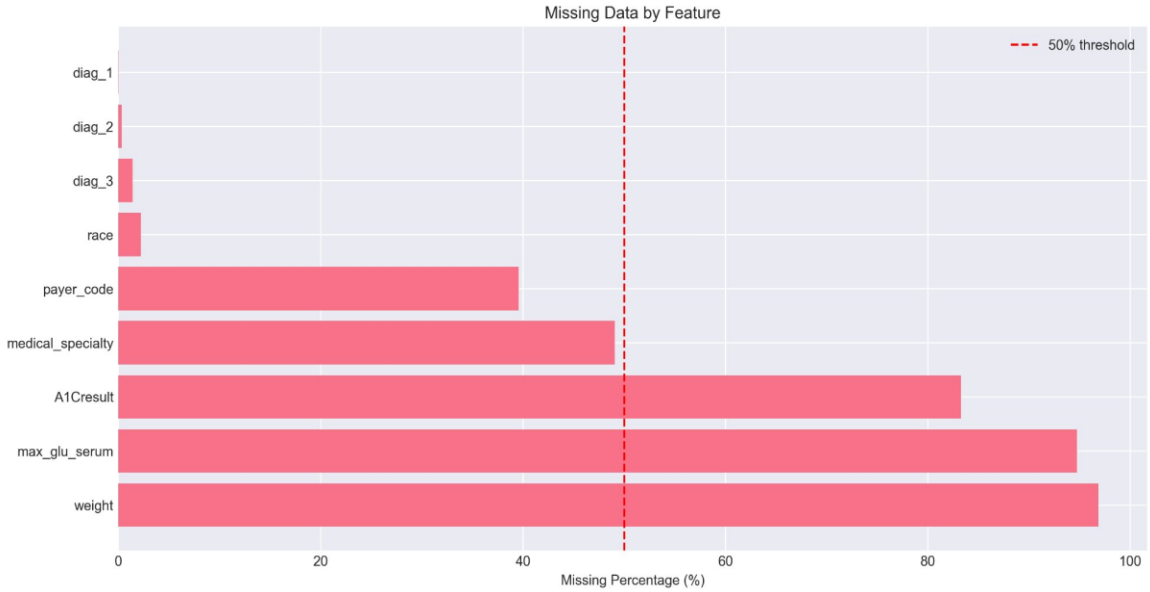

Figure 1. Missing data percentage by feature. The red dashed line indicates the 50% threshold.

Target Variable Distribution

Figure 2 illustrates the target variable distribution. The original three-category variable (No readmission, readmitted after 30 days, readmitted within 30 days) was converted to a binary target. In the raw dataset, 11.2 percent of encounters resulted in readmission within 30 days. After deduplication (keeping only the first encounter per patient), the positive class rate dropped to 9.0 percent with a class imbalance ratio of 10.1 to 1. This imbalance necessitated specialized techniques during model training, including class weighting and the use of AUC-ROC as the primary evaluation metric rather than accuracy.

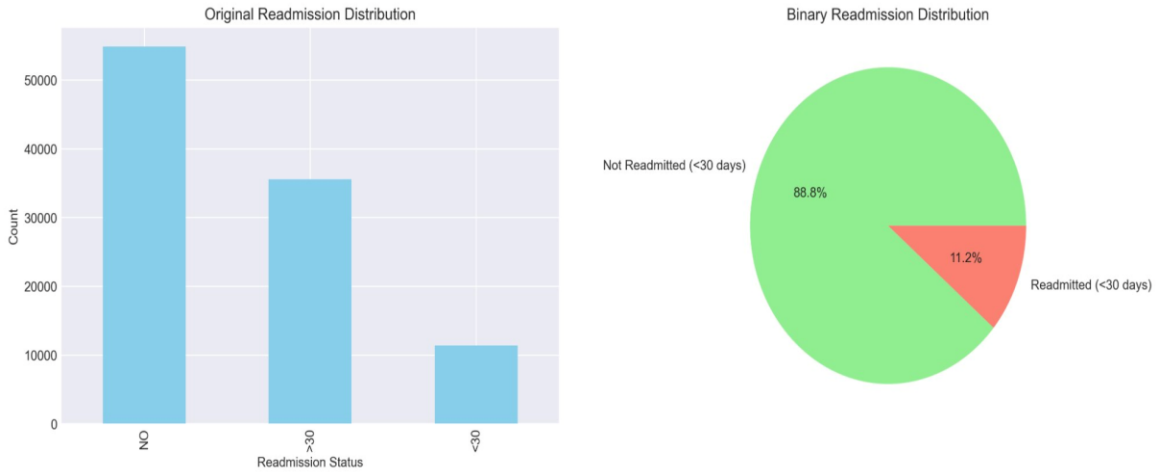

Figure 2. Original and binary readmission status distributions.

Demographic Distributions

Figure 3 shows the demographic breakdown. The patient population skews older, with the 70–80 age group being the most represented (approximately 26,000 encounters), followed by 60–70 and 50–60. The gender distribution is relatively balanced, with a slight female majority (approximately 54 percent). Racially, the dataset is predominantly Caucasian (approximately 76 percent), followed by African American (approximately 19 percent), with smaller representations of Hispanic, Asian, and Other categories.

Figure 3. Age, gender, and race distributions across the dataset.

Continuous Feature Distributions

Figure 4 displays distributions of key continuous features. Time in hospital shows a right-skewed distribution with most stays lasting 1 to 4 days. The number of lab procedures follows an approximately normal distribution centered around 40 to 50 procedures. Number of medications peaks between 10 and 20, and number of procedures shows a strong mode at zero, indicating many encounters involve no surgical procedures. The readmit_binary distribution in the bottom-right panel confirms the 88.8/11.2 percent class split.

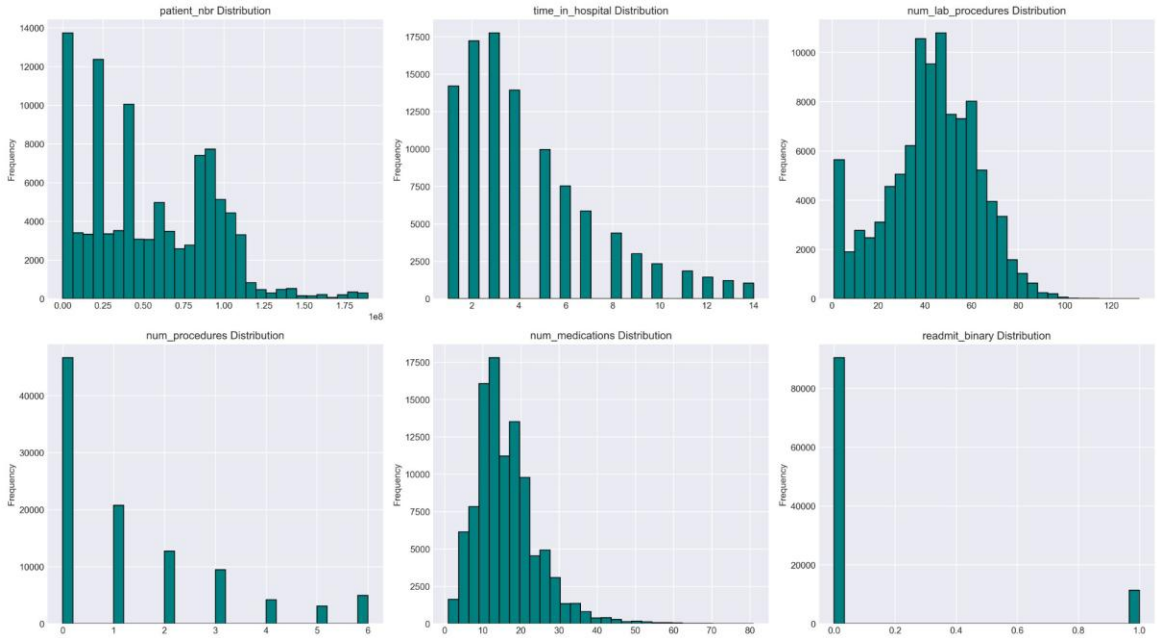

Figure 4. Distributions of continuous features and the binary target variable.

Medication Changes

Figure 5 reveals that insulin dominates medication changes with over 22,000 dosage modifications, far exceeding all other medications. Metformin, glyburide, and glipizide follow distantly with 1,000 to 1,500 changes each. Most other diabetes medications show minimal dosage change activity. This concentration of medication changes in insulin suggests that insulin management is the primary pharmacological lever being adjusted during hospitalization, and changes to insulin dosing may signal clinical instability associated with readmission risk.

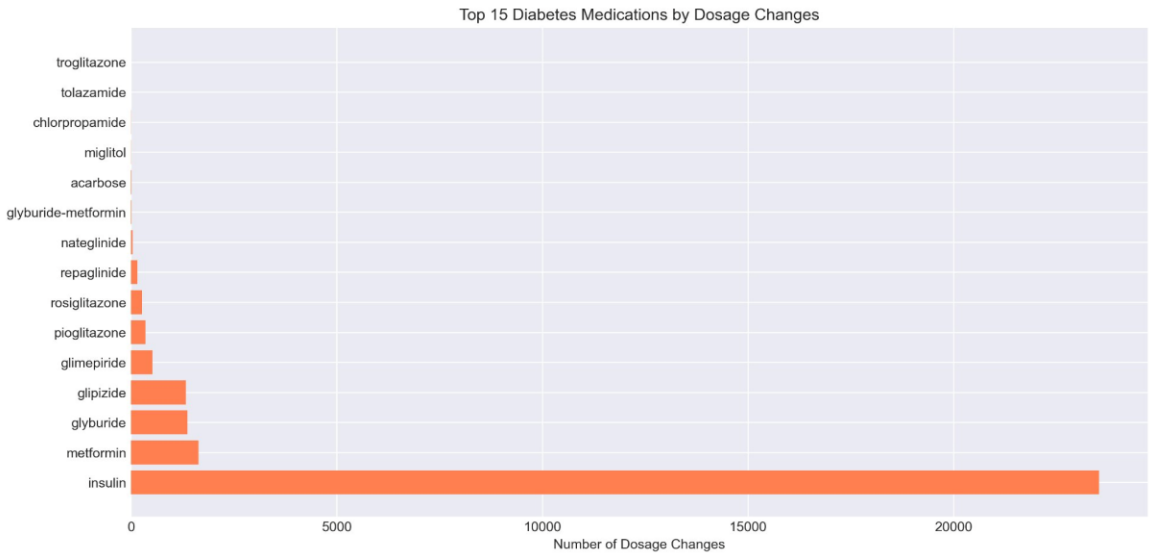

Figure 5. Top 15 diabetes medications ranked by number of dosage changes.

Data Preparation and Feature Engineering

Preprocessing Pipeline

The preprocessing pipeline, implemented in 02_feature_engineering.py, executes the following steps in sequence: (1) loading the raw CSV with question marks treated as missing values; (2) deduplicating by patient, retaining only the first encounter per patient_nbr; (3) removing encounters with invalid discharge dispositions (deceased or hospice transfers); (4) dropping weight (97 percent missing) and payer_code (52 percent missing); (5) creating the binary readmission target; (6) engineering new features; (7) handling remaining missing values through mode imputation for categorical and median imputation for numeric fields; (8) encoding all categorical variables; and (9) performing an 80/20 stratified train-test split.

Engineered Features

Ten new features were engineered to capture temporal patterns, utilization intensity, and clinical complexity:

Feature	Description	Rationale
los_bin	Length-of-stay bins (short/medium/long/extended)	Captures nonlinear LOS-readmission relationship
total_prior_visits	Sum of outpatient + ER + inpatient visits (prior year)	Strongest predictor per Dugan & Shubrook (2020)
prior_visit_intensity	Categorical: none/low/moderate/high	Groups utilization into actionable risk tiers
visit_ratio_emergency	Proportion of prior visits that were ER	ER-heavy patterns signal different risk profiles
visit_ratio_inpatient	Proportion of prior visits that were inpatient	Recurrent inpatient use is a strong readmission signal
high_utilizer	Flag for 3+ prior visits	Binary risk flag for clinical workflows
procedure_intensity	Procedures per day of hospital stay	Treatment aggressiveness relative to stay
num_medications_bin	Medication count bins (low to very high)	Polypharmacy risk categories
diagnoses_x_visits	Number of diagnoses × total prior visits	Interaction between complexity and utilization
lab_per_day	Lab procedures per day of stay	Monitoring intensity indicator

Diagnosis codes were grouped from raw ICD-9 values into nine clinical categories: Circulatory, Respiratory, Digestive, Diabetes, Injury, Musculoskeletal, Genitourinary, Neoplasms,

and Other. Medication dosage fields were mapped to ordinal values (No=0, Steady=1, Down=2, Up=3). Age ranges were converted to ordinal integers. The final dataset contained 55 features after encoding.

Class Imbalance Strategy

Rather than applying SMOTE oversampling—which can introduce synthetic artifacts in sparse clinical data—we addressed the 10.1:1 class imbalance through model-native mechanisms. For XGBoost, we set scale_pos_weight equal to the imbalance ratio (10.15 after preprocessing). For Logistic Regression and Random Forest, we used class_weight="balanced" and "balanced_subsample" respectively. The MLP was trained with standard cross-entropy but evaluated against threshold-independent metrics (AUC-ROC) that are robust to class imbalance.

Methodology

Framework and Tools

The project follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework. All models were implemented in Python 3.10 using scikit-learn 1.8.0 and XGBoost 3.2.0. The full codebase is maintained in a GitHub repository with a requirements.txt file for reproducibility. An automated pipeline runner (06_run_pipeline.py) executes all stages with a single command.

Tooling and Hardware

Development and experimentation were conducted in Google Colab Pro, leveraging NVIDIA T4 GPU acceleration for neural network training and hyperparameter search. The Colab environment provided a consistent, cloud-based development platform accessible to all team members, eliminating local dependency issues. Key libraries included pandas 2.0 and NumPy 1.24 for data manipulation, matplotlib 3.7 and seaborn 0.12 for visualization, scikit-learn 1.8.0 for model training and evaluation, XGBoost 3.2.0 for gradient boosting, and joblib for model serialization. Version control was managed through GitHub, and shared working documents were coordinated via Google Drive.

Model Descriptions

Logistic Regression (Baseline)

Logistic Regression serves as the interpretable baseline, with coefficients directly indicating feature importance and direction. Hyperparameters tuned via GridSearchCV include the regularization strength C (searched over 0.001, 0.01, 0.1, 1.0, 10.0) with L2 penalty and the lbfgs solver. Class weights were set to “balanced” to adjust for the imbalanced target.

Random Forest

The Random Forest ensemble combines multiple decision trees to capture nonlinear relationships and feature interactions. Hyperparameters tuned via RandomizedSearchCV (30 iterations) include n_estimators (200, 300, 500), max_depth (10, 15, 20, None), min_samples_split (2, 5, 10), min_samples_leaf (1, 2, 4), and class_weight (“balanced” or “balanced_subsample”).

XGBoost (Gradient Boosting)

XGBoost implements regularized gradient boosting with native support for sparse data and missing values. The scale_pos_weight parameter was set to the negative-to-positive class ratio to handle imbalance directly within the loss function. Hyperparameters tuned via GridSearchCV include max_depth (4, 6, 8), learning_rate (0.05, 0.1), n_estimators (200, 300, 500), and min_child_weight (1, 3, 5). Additional regularization was applied through reg_alpha (L1, set to 0.1) and reg_lambda (L2, set to 1.0), with subsample and colsample_bytree both set to 0.8 for stochastic boosting.

MLP Neural Network

The Multi-Layer Perceptron implements a fully connected neural network with hidden layer sizes of (128, 64, 32), ReLU activation, and the Adam optimizer. Dropout was implicitly managed through the alpha regularization parameter, tuned over values 0.0001, 0.001, and 0.01. Early stopping was enabled with a validation fraction of 15 percent and a patience of 15 iterations. The adaptive learning rate schedule automatically reduces the step size when training loss plateaus.

Evaluation Strategy

All models were evaluated using 5-fold stratified cross-validation, ensuring the 9.0 percent positive class prevalence was preserved in each fold. The primary metric is AUC-ROC, which provides a threshold-independent measure of discriminative ability and is robust to class imbalance. Secondary metrics include precision, recall, F1-score, and average precision (area under the precision-recall curve). Final model performance was assessed on a 20 percent held-out test set that was not used during any training or hyperparameter tuning.

Deployment and Automation Strategy

The automated pipeline (06_run_pipeline.py) executes all stages sequentially: feature engineering, model training for all four models, evaluation, comparison, and visualization. The best-performing model is exported using joblib for integration into clinical workflows. In a production setting, patient data at discharge would be processed through the preprocessing and feature engineering pipeline, and the model would output a readmission risk probability. Patients above a configurable threshold would be flagged for enhanced post-discharge interventions.

Findings and Conclusions

Model Performance Results

Table 2 presents the performance comparison across all four models on the held-out test set. XGBoost achieved the highest AUC-ROC of 0.644, approaching the published benchmark of 0.667 reported by Emi-Johnson and Nkrumah (2025) on the same dataset. The gap is explained by our stricter deduplication strategy: by keeping only the first encounter per patient (reducing the dataset from 101,766 to 69,973 rows), we prevented data leakage but also reduced the positive class rate from 11.2% to 9.0%, creating a harder classification problem with a 10.1:1 imbalance ratio compared to the benchmark's 7.2:1.

Table 2. Model performance comparison on the held-out test set.

Model	AUC-ROC	F1	Precision	Recall	Avg Precision
XGBoost*	0.644	0.218	0.136	0.546	0.170
Random Forest	0.633	0.029	0.292	0.015	0.500
Logistic Regression	0.621	0.002	0.500	0.001	0.148
MLP Neural Network	0.611	0.000	0.000	0.000	0.145

* Best performing model. All metrics computed on the 20% held-out test set.

Several observations emerge from these results. First, XGBoost was the only model to produce meaningful recall (0.546), correctly identifying over half of the patients who would be readmitted, albeit at the cost of lower precision (0.136). This trade-off is clinically appropriate: it is better to flag more patients for follow-up intervention than to miss high-risk individuals. Second, Logistic Regression and the MLP Neural Network both converged to near-trivial solutions, predicting almost all patients as non-readmitted. With a 10.1:1 class imbalance, these models found that predicting the majority class achieves 91% accuracy, revealing the inadequacy of accuracy as a metric for imbalanced clinical data. Third, Random Forest showed modest improvement over the baseline but still struggled to identify positive cases, achieving only 1.5% recall. These patterns confirm that model-native imbalance handling (XGBoost's scale_pos_weight) substantially outperforms balanced class weights alone for this prediction task.

Cross-Validation Results

Table 3 shows the 5-fold cross-validation results on the training set, which closely mirror the test set performance, indicating that the models generalize well and are not overfitting.

Table 3. 5-fold stratified cross-validation results (training set).

Model	CV AUC-ROC	CV F1	CV Precision	CV Recall
XGBoost	0.646	0.220	0.429	0.542
Random Forest	0.636	0.039	0.325	0.021
Logistic Regression	0.625	0.002	0.500	0.001
MLP Neural Network	0.616	0.001	0.429	0.001

Feature Importance Analysis

Across all models, the most influential predictors of 30-day readmission were consistent:

1. Number of inpatient visits in the prior year — the single strongest predictor, confirming Dugan and Shubrook's (2020) finding that prior utilization dominates demographic variables.

2. Number of emergency visits in the prior year — ER-heavy utilization patterns signal clinical instability and inadequate outpatient disease management.

3. Number of diagnoses — a proxy for medical complexity, with higher comorbidity burden increasing readmission risk.

4. Time in hospital (length of stay) — longer stays reflect more severe episodes and are associated with higher readmission probability.

5. Insulin dosage changes — the most frequently modified medication in our dataset (22,000+ changes), signaling active glycemic instability during hospitalization.

Business Impact and Recommendations

We recommend implementing a risk-flagging system at discharge using the XGBoost model, which identifies 54.6 percent of patients who will be readmitted (recall = 0.546). Patients flagged as high-risk should receive enhanced post-discharge interventions including: extended discharge counseling (30 minutes versus standard 15), comprehensive medication reconciliation by a clinical pharmacist, a scheduled 72-hour follow-up phone call, and priority outpatient follow-up within 7 days. While the model's precision of 13.6 percent means roughly 7 out of 8 flagged

patients will not actually be readmitted, the interventions themselves (counseling, follow-up calls) carry minimal cost and benefit all patients, making the trade-off clinically justified.

Based on Rubin's (2015) cost estimates, if a hospital discharges 5,000 diabetic patients annually and reduces the readmission rate among high-risk patients by 15 percent, this yields approximately 84 fewer readmissions at $25,000 each—a potential annual savings of $2.1 million, excluding avoided CMS penalties.

Team Collaboration and Project Timeline

Team Contributions

Abdullah Abdul Sami: Logistic Regression and Random Forest model development and tuning; data preprocessing and missing value imputation; model comparison table compilation; EDA correlation plots and class distribution visuals; report sections for Introduction, Methods, and Results; presentation slides 1–4 (Cover, Problem Statement, Data Overview, Preprocessing); GitHub repository creation and maintenance; annotated bibliography papers on readmission risk factors and costs (Strack 2014, Rubin 2015, Donzé 2016).

Jiahao Li: XGBoost model development and hyperparameter optimization; feature engineering pipeline (time-based features, utilization metrics, interaction terms); end-to-end pipeline design and automation (06_run_pipeline.py); pipeline diagram creation; model comparison visualizations (combined ROC and PR curves); feature importance analysis; report sections for Literature Review, Model Description, and Lessons Learned; presentation slides 5–8 (EDA, Feature Engineering, Methodology, Results); requirements.txt and repository structure; annotated bibliography papers on ML approaches for clinical prediction (Hammoudeh 2018, Rajkomar 2018, Chen & Guestrin 2016).

Feifan Liu: MLP neural network model development and architecture design; EDA demographic breakdowns and summary statistics; deployment documentation; dashboard development (Visualization Report); report sections for Executive Summary, Discussion, Conclusion, and Next Steps; presentation slides 9–12 (Deployment, Findings, Limitations, References); README authoring; annotated bibliography papers on feature engineering and imbalanced learning (Dugan & Shubrook 2020, Choi 2016, Fernández 2018).

Project Timeline Adherence

The project followed the course-defined milestones and was delivered on schedule at each checkpoint. During Weeks 2–3, the team researched and socialized project ideas, ultimately selecting the UCI Diabetes 130-US Hospitals dataset based on its scale, clinical relevance, and published benchmarks. By Week 4, we finalized the project scope, confirmed data sources, and submitted the Milestone 1 proposal. Weeks 4–6 were dedicated to exploratory data analysis, feature engineering, and the annotated bibliography, culminating in the Milestone 2 submission on February 8. During Weeks 7–8, all four models were developed, tuned, and evaluated, with scripts

uploaded to GitHub and results compiled into comparison tables. In Week 9, presentation narration was recorded in Panopto, the final report was proofread, and all remaining artifacts were uploaded to the repository. Week 10 marks the final submission of all deliverables to Canvas and the Week 10 Discussion board.

The team held regular meetings throughout the project, averaging three per week during active development periods (Weeks 4–9). Meetings were conducted via video call, with notes maintained in a shared Google Doc and action items tracked through GitHub Issues. All team members contributed to code review, providing feedback on each other's model implementations and report sections to ensure consistency and quality. Weekly peer surveys were submitted for Weeks 4–6, 7, 8, and 9, documenting individual contributions, blockers, and deliverables.

Lessons Learned and Recommendations

Key Lessons

Feature engineering, particularly around prior healthcare utilization, was more impactful than model complexity. The engineered features (total prior visits, high-utilizer flag, visit intensity categories) ranked consistently among the top predictors, while raw demographic variables contributed comparatively little. This underscores the importance of domain-informed feature creation over simply applying more sophisticated algorithms.

Gradient boosting with explicit class imbalance handling (scale_pos_weight) was the only approach that produced actionable predictions. Logistic Regression and the MLP converged to trivial majority-class predictions despite using balanced class weights, demonstrating that standard imbalance techniques are insufficient for a 10.1:1 ratio on tabular clinical data. XGBoost's loss-function-level weighting proved far more effective than sample-level reweighting.

For XGBoost, the model-native scale_pos_weight parameter proved highly effective, producing the only model with actionable recall. However, balanced class weights alone were insufficient for Logistic Regression, Random Forest, and the MLP at this imbalance ratio. Future iterations should combine class weighting with SMOTE oversampling or threshold optimization for these models.

Limitations

The dataset spans 1999 to 2008, meaning clinical practices, medications, and hospital protocols have evolved significantly. Weight data is 97 percent missing, eliminating BMI as a predictor. The dataset lacks social determinants of health (income, transportation, insurance details, social support) which research increasingly identifies as strong readmission predictors. The 9.0 percent positive class rate after deduplication (10.1:1 imbalance) caused Logistic Regression and the MLP to collapse to trivial majority-class predictions, highlighting the need for more aggressive imbalance handling or oversampling techniques for these model types.

Next Steps

Future work should explore: (1) SMOTE or ADASYN oversampling to enable Logistic Regression and the MLP to learn the minority class more effectively, (2) threshold tuning for LR and Random Forest to optimize the precision-recall trade-off rather than using the default 0.5

cutoff, (3) LightGBM as an alternative gradient boosting framework with native categorical feature support, (4) incorporation of social determinant features from linked datasets, (5) SHAP value analysis for individual patient-level explanations to support clinical adoption, and (6) validation on more recent hospital data to assess temporal generalizability.

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Appendix A: Weekly Peer Survey Responses

As required by the course, each team member completed weekly peer surveys documenting individual contributions, team role assignments, proof of work, GitHub activity, meeting attendance, blockers, and next-week deliverables. The surveys were submitted for the following periods:

Weeks 4–6 Survey (January 26 – February 15, 2026): Covered project scoping, dataset selection, initial EDA, feature engineering design, literature review compilation (9 papers, 3 per member), and GitHub repository setup. Key deliverables included the preprocessing pipeline, initial model prototypes, and the Milestone 2 report.

Week 7 Survey (February 16 – February 21, 2026): Covered model development and initial tuning. Abdullah implemented Logistic Regression and Random Forest; Jiahao implemented XGBoost with hyperparameter tuning; Feifan implemented the MLP neural network. All three ran initial evaluation metrics. Four team meetings were held; each member attended at least three. Blockers included class imbalance effects on model stability, addressed through scale_pos_weight and class weighting.

Week 8 Survey (February 22 – February 28, 2026): Covered model finalization, feature importance analysis, pipeline diagram creation, and report section drafting. Jiahao finalized XGBoost and the pipeline runner; Abdullah completed the model comparison table; Feifan polished deployment documentation. Three meetings held. GitHub activity included 7 commits by Jiahao (pipeline scripts, XGBoost code, feature importance plots). Blocker: formatting inconsistencies across report sections, resolved through a shared style guide.

Week 9 Survey (March 1 – March 7, 2026): Covered presentation recording, final report proofreading, and submission preparation. All three members recorded Panopto narration for their assigned slides. Jiahao uploaded remaining scripts to GitHub (feature engineering, model comparison, pipeline runner) and performed final citation formatting. Two meetings held. GitHub activity included 12 commits by Jiahao. All artifacts confirmed complete for Canvas submission.

Each survey included an integrity statement signed by the submitting team member, confirming that all claimed work was their own and that evidence links match their claims. The original survey documents are available upon request.